MILITARY SPECIFICATION

INTERCONNECTING GROUP, ON-143(V)6/USQ

This amendment forms a part of Military Specification MIL-I-29502(EC), dated 21 April 1986, and is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

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Preparing activity:
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4. Voice loop test
5. Data loop test
6. Control panel test

i. Provide the necessary low-level outputs for the teletypewriter (TTY) and the typing reperforator.

j. By means of a front-mounted control panel, allow the operator to select mode of operation, enter operating parameters, and verify and enter data by touching the appropriate indicator. For ease of operation, the front-mounted control panel shall be capable of being remotely located.

The IG shall be a standard 19-inch (in.) rack-mountable unit with an enclosed chassis. All controls and indicators necessary for operation shall be located on the front panel, although it shall be possible to remotely locate the control panel. A cover shall be provided for the IG when the control panel is in a remote location. All input, output, and power connectors shall be mounted on the rear panel. The IG front panel and chassis shall be mounted on slide rails to allow the IG to be extended from the case for access to the card cages and internal circuitry. The IG shall be broken down into the five basic areas specified in k through o:

k. Front panel, rear panel, and case
l. RED card cage
m. BLACK card cage
n. RED/BLACK interface
o. Power supplies

FIGURE 1 is an interface block diagram of the IG and system equipment. FIGURE 2 is an overall functional block diagram of the IG.

3.3.1.1 Front panel, rear panel, and case. The front panel shall contain the operating controls and indicators and the control panel. An internally-mounted fan and radio-frequency interference (RFI) attenuation air filter shall be used to cool the IG. When the front panel is mated with the housing, an RFI gasket on the front panel shall ensure RFI integrity. The front end chassis shall be capable of being extended by first loosening knurled fasteners at each corner and pulling outward on the handles on both sides of the front panel until the detents in the side rails engage. With the unit so extended, the RED and BLACK card cages, power supplies, and internal circuits shall be accessible. Flat cables with plug-in connectors shall provide the interface between the rear panel connectors and the IG circuitry.

3.3.1.2 RED card cage. The RED card cage shall be mounted to the chassis and shall be extended out of the case with the front panel. The RED card cage shall contain the 13 circuit cards that perform the various classified system functions. The RED card cage shall contain the cards specified in a through d:

a. One central processing unit (CPU)
b. Four random access memory (RAM)
c. Four read only memory (ROM)
d. Four dynamically adaptive receiver transmitter (DART):
   1. One vocoder DART
   2. One crypto DART
   3. One CPTTY DART
   4. One data link DART

Circuitry from the RED card cage to the rear panel shall be in the form of flat multiconductor ribbon cables. The RED card cage end of the ribbon cables shall terminate with 40-pin male strip plugs. The rear panel end of the ribbon cables shall terminate with round multipin connectors on the rear panel. The flexibility of the ribbon cables shall allow the chassis to be extended on the slide rails.

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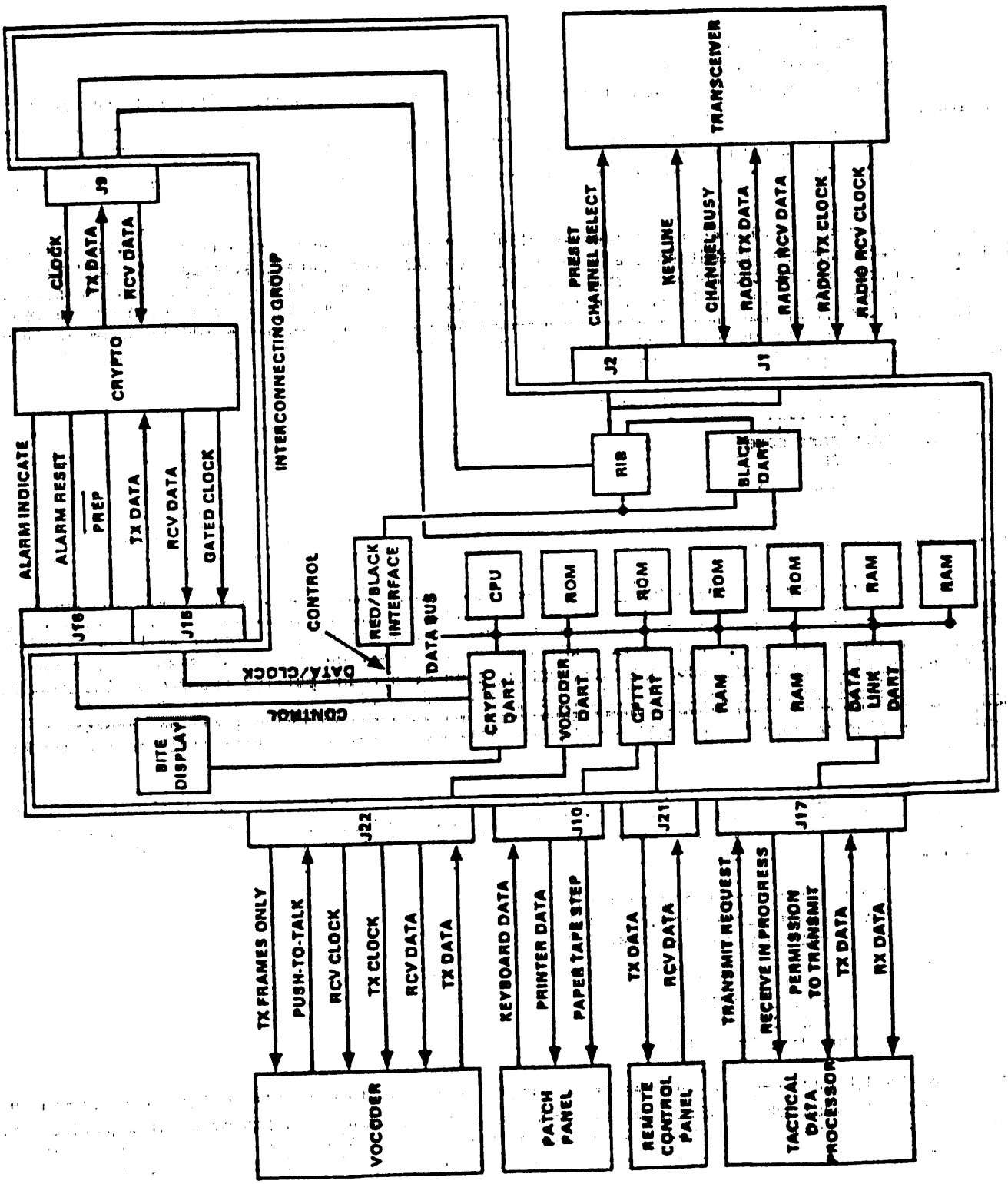


FIGURE 1. Detailed interface block diagram.

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3.3.2.2.1 Application of BLACK power. The IG shall be capable of direct application of unfiltered (BLACK) power without compromising emanations being present on the power line.

3.3.2.3 Strappable options. Provision shall be made for the selection of the options specified in a and b by backplane strapping (see 6.2):

a. Exclusion of SSIXS function. This option is appropriate for equipment to be installed onboard surface ships which will not require SSIXS capability.
b. Selection of TTY data rates 75 bits per second (bps), 300 bps, 1200 bps or 2400 bps

3.3.3 Major component list. The IG shall consist of the components specified in TABLE II. The component parts shall be fully interoperable and interchangeable with previously manufactured IG components to facilitate interoperability, interchangeability, training, and logistics.

TABLE II. Major component list.

COMPONENT	DWG No.
Interconnecting Group (1)	55910-0104160
Panel Assembly, Front (1A1)	55910-0104146
Control Indicator, C-10582/USQ (1A1A1)	55910-0104153
Panel Assembly, Rear (1A2)	55910-0104645
Cage Assembly, BLACK Circuit Module (1A3)	55910-0104152
Cage Assembly, RED Circuit Module (1A4)	55910-0104151
Panel, Power Distribution (1A5)	55910-0104642
Circuit Card, RED/BLACK Interface (1A6)	55910-0104604
Circuit Card, BLACK DART (1A3A3)	55910-0104137
Circuit Card, RIB (1A3A6)	55910-0104140
Circuit Card, Vocoder DART (1A4A32)	55910-0104137
Circuit Card, CPTTY DART (1A4A30)	55910-0104137
Circuit Card, Data Link DART (1A4A29)	55910-0104137
Circuit Card, Crypto DART (1A4A28)	55910-0104137
Circuit Card, RAM (1A4A20) (C)	89199-0011101
Circuit Card, RAM (1A4A18) (A)	89199-0011101
Circuit Card, RAM (1A4A17) (9)	89199-0011101
Circuit Card, RAM (1A4A16) (8)	89199-0011101
Circuit Card, ROM (1A4A15) (3)	89199-0011097
Circuit Card, ROM (1A4A14) (2)	89199-0011097
Circuit Card, ROM (1A4A13) (1)	89199-0011097
Circuit Card, ROM (1A4A12) (0)	89199-0011097
Circuit Card, CPU (1A4A11)	89199-0008122
Power Supply (1PS1)	55910-0104157-1
Power Supply (1PS2)	55910-0104157-2
Cable Assembly (1W1)	55910-0105714
Cable Assembly (1W2)	55910-0105715
Cable Assembly (1W3)	55910-0105716
Cable Assembly (1W4)	55910-0105717
Cable Assembly (1W5)	55910-0105718
Cable Assembly (1W6)	55910-0105719
Cable Assembly (1W7)	55910-0105720
Cable Assembly (1W8)	55910-0105721
Cable Assembly (1W9)	55910-0105722
Cable Assembly (1W10)	55910-0105723
Cable Assembly (1W11)	55910-0105724
Cable Assembly (1W12)	55910-0105725
Cable Assembly (1W13)	55910-0105726
Cable Assembly (1W14)	55910-0105727

3.3.4 Mechanical design. Mechanical design shall be as specified in 3.3.4.1 through 3.3.4.5.5.

3.3.4.1 Enclosures. The IG enclosures shall conform to the Enclosures paragraph of MIL-E-16400 and 3.3.4.2.

3.3.4.2 Degree of enclosure. The IG enclosure shall be dripproof (45 degrees) in accordance with MIL-STD-108.

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3.3.4.3 Equipment slide and tilt mechanisms. The equipment shall have slide mechanisms which shall be of suitably rugged construction to support the drawer in the withdrawn position. The slide mechanism shall slide with a minimum of sliding friction without a tendency to bind. The slide mechanism shall be constructed of corrosion-resistant material, such as corrosion-resistant steel. Guide support shall be provided between the rear of each chassis and the enclosure to prevent possible damage, due to shock, to the chassis slides.

3.3.4.4 Shock mounting. All slide-mounted equipment shall include shock pins at the rear of the unit.

3.3.4.5 Controls, indicators, and panel layouts. The design of controls and indicators, and their arrangement on operator and maintainer panels, shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the Controls, indicators, and panel layout paragraph of MIL-E-16400. The color coding and marking of pin jacks and other test points shall be in accordance with MIL-STD-415. Abbreviations used on panels shall be in accordance with MIL-STD-12.

3.3.4.5.1 Front panel layout. Front panel layout of cabinets and drawers shall take into consideration sequencing procedures to facilitate performance by Navy electronics technicians, second class (ET-2) for set up, operation, and maintenance. All controls and metering necessary for normal operation of the IG shall be provided on the IG front panel.

3.3.4.5.2 Alarm lamps. Lamps used on front panels for alarms and other purposes shall conform to the requirements of the Transilluminated Displays paragraph of MIL-STD-1472. A capability for remote indication of the status of the alarm lamps shall be provided. Legend lights shall be used in preference to simple indicator lights except where design considerations demand that simple indicators be used. Lamps used in legend switches shall conform to the requirements of the Legend Switches paragraph of MIL-STD-1472. Legend switches shall be used in preference to toggle switches except where design considerations demand that toggle switches be used. Lamp testing shall be in accordance with the Lamp Testing paragraph of MIL-STD-1472.

3.3.4.5.3 Protection from improper operation. The IG design shall include features to protect the equipment from improper operational procedures and practices.

3.3.4.5.4 Chain reaction failures. To the extent practicable, the IG protective design shall inhibit chain reaction failures in which a failure in one component results in damage to other components.

3.3.4.5.5 Locking devices. Locking devices shall be provided on all critical internal adjustment controls in order to prevent the need for readjustment of controls after sustained periods of shock and vibration.

3.3.5 Airborne and structureborne noise. The IG shall conform to the requirements of MIL-STD-740 for Grade A, Type 3 equipment and as specified herein. The maximum allowable sound pressure level for airborne noise shall be 60 decibels amperes measured at 1 meter distance from the equipment. The structureborne noise criteria shall be 65 decibels (dB) for one third wave band frequencies up to 800 Hz. Both the airborne and structureborne noise criteria shall be applicable when the equipment is operating normally.

3.3.6 Thermal design. The thermal design for the IG shall be as specified in 3.3.6.1 and 3.3.6.2 and in accordance with the Thermal design paragraph of MIL-E-16400. Air cooling shall be used. The design of the equipment shall allow installation in open racks.

3.3.6.1 Heat removal. The heat removal method shall remove at least 80 percent of the heat generated within the equipment. MIL-HDBK-251 may be used as guidance in the design and selection of the heat removal method.

3.3.6.2 Cooling during maintenance. Cooling during maintenance shall conform to the Cooling during maintenance paragraph of MIL-E-16400.

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